OPEN-SOURCE ONLINE TOOLS (READY TO USE)

Below are **popular, free, open-source tools** you can deploy online for a **Sales & Profit Analyzer**.

1.Metabase (Highly Recommended)

- **Type:** Open-source BI & analytics
  Best for: Sales & profit dashboards
  AI support: Via SQL + ML integration

Features:

- Upload CSV / connect database
- Profit & sales dashboards
- Filters (date, product, region)
- Easy charts & reports

Tech:

- PostgreSQL / MySQL / CSV
- Docker supported

Use case:

- “Small shop owner views monthly sales & profit trends online”

2. Apache Superset

- **Type:** Advanced open-source analytics platform
  Best for: Large datasets & AI-ready dashboards

Features:

- Interactive dashboards
- SQL Lab for analytics
- ML integration (Python backend)
- Role-based access

Ideal for:

- AI-powered forecasting
- Multi-branch business analysis

3. ERPNext (All-in-One Open Source ERP)

- **Type:** Business management system
  Best for: Real business deployment

Features:

- Sales, inventory, accounting
- Profit & loss statements
- Customer & product analytics
- REST APIs for AI integration

AI Add-on:

- Export data → apply ML models
- Plug AI insights back into ERP

4. Streamlit (AI Dashboard Builder)

- **Type:** Python open-source framework
  Best for: AI-based analytics tools

Features:

- Upload sales CSV
- Auto profit calculation
- ML forecasting (LSTM / Prophet)
- Runs online (Streamlit Cloud)

Example use:

- AI-powered sales & profit analyzer in one Python app

5. Odoo Community Edition

- **Type:** Open-source business suite
  Best for: Sales + accounting analytics

Features:

- Sales & invoice tracking
- Profit analysis
- Open APIs
- AI-ready architecture

AI STACK (Open Source)

Purpose	Tool
Data processing	Pandas, NumPy
ML models	Scikit-learn
Forecasting	Prophet, LSTM
Visualization	Plotly
Backend	FastAPI
Auth	JWT

- **RECOMMENDED COMBINATION (BEST PRACTICE)**

Academic / Mini Project

- Streamlit + Pandas + Scikit-learn

Small Business (Real Use)

- ERPNext + Metabase + AI API

Advanced AI System

- FastAPI + Superset + ML Models

How to create csv file

- Creating a **CSV (Comma-Separated Values) file** is simple. Here are the **most common and practical ways**, from beginner-friendly to programmatic.

METHOD 1: Using Microsoft Excel (Easiest)

- Open **Microsoft Excel**
- Enter data in rows and columns

- Example:

Date	Product	Sales	Cost
2024-01-01	Soap	5000	3200
2024-01-02	Shampoo	7000	4500

- Click **File → Save As**
- Choose location
- Select **Save as type:** CSV (Comma delimited) (*.csv)
- Click **Save**
- Click **Yes** if Excel shows a warning
- File saved as: sales.csv

- **METHOD 2: Using Google Sheets (Online & Free)**
- Open **Google Sheets**
- Enter your data
- Click **File → Download**
- Select **Comma Separated Values (.csv)**
- ✓ Works fully online

- **METHOD 3: Using Notepad (Manual)**
- Open **Notepad**
- Type data with commas:

  Date,Product,Sales,Cost

  2024-01-01,Soap,5000,3200

  2024-01-02,Shampoo,7000,4500

  2024-01-03,Soap,6200,3500
- Click **File → Save As**
- File name: sales.csv
- Save as type: **All Files**
- Encoding: **UTF-8**
- Click **Save**

- **METHOD 4: Using Python (Recommended for AI Projects)**

```
import pandas as pd

data = {
    "date": ["2024-01-01", "2024-01-02", "2024-01-03"],
    "product": ["Soap", "Shampoo", "Soap"],
    "sales": [5000, 7000, 6200],
    "cost": [3200, 4500, 3500]
}

df = pd.DataFrame(data)
df.to_csv("sales.csv", index=False)

print("CSV file created successfully")
```

Output file: sales.csv

METHOD 5: Using Online CSV Generator Tools

- Free tools:
- TableConvert
- CSVJSON
- ConvertCSV
- ✓ No installation needed
  ✓ Good for quick datasets